

MUHAMMAD HASIF BIN MOHD FAISAL

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [ORCID](#)

PROFESSIONAL SUMMARY

Applied Artificial Intelligence undergraduate at the Singapore Institute of Technology (expected 2028) with experience delivering data pipelines, machine-learning models, LLM applications, REST APIs, and analytical interfaces across maritime research, consulting, public-sector operations, and cybersecurity. Strong in Python, SQL, scikit-learn, Django REST Framework, FastAPI, React, relational databases, automation, testing, and stakeholder communication.

TECHNICAL SKILLS

Programming and Data: Python, SQL, JavaScript/TypeScript, R; pandas, NumPy, ETL, batch processing, data cleaning and validation, relational modelling, exploratory data analysis, feature engineering

Machine Learning and Analytics: scikit-learn, imbalanced-learn, XGBoost, LightGBM, ARIMA/SARIMA, classification, forecasting, SMOTE, cross-validation, temporal validation, threshold optimisation, ROC-AUC, precision, recall, F1, RMSE

Applied AI and Applications: Gemini, Qwen, Llama, LangChain, function calling, embeddings, semantic search; Django, FastAPI, React, REST APIs, PostgreSQL/pgvector, Microsoft SQL Server, OAuth 2.0, JWT

Visualisation and Delivery: Power BI, Streamlit, ArcGIS, Excel/VBA, UiPath RPA, Git, Docker, pytest, GCP, Oracle Cloud; dashboards, workflow automation, technical documentation

PROFESSIONAL EXPERIENCE

Assistant Research and Full-Stack Developer

Mar 2026 - Present

Singapore Institute of Technology - Future Ship & Systems Design Project, Singapore

Private Repositories: [Backend Link](#) | [Frontend Link](#)

- Develop an end-to-end maritime data and analytics platform that transforms electric-harbour-craft SCADA, sensor, and alarm data into near-real-time vessel-health monitoring, explainable fault diagnosis, and engineering decision support.
- Drive full-stack delivery for the current research phase across Python, Django REST Framework, SQL Server, React, and Vite, covering multi-vessel data models, ETL and historical-replay pipelines, authenticated REST APIs, configurable FMECA workflows, dashboards, and interactive fault-evidence views.
- Engineer reliable ingestion and transformation for high-volume battery and propulsion telemetry, handling sparse signals, duplicates, missing/sentinel values, subsystem mapping, and batched persistence; translate engineering and FMECA requirements into traceable fault logic, testing, documentation, and iterative product delivery, validating workflows on multi-million-row datasets and improving local replay performance by 87%.
- Collaborate with maritime researchers and engineers to translate operational requirements into data models, APIs, diagnostic workflows, and analytical interfaces while balancing data quality, traceability, performance, and usability.

Assistant Research and Software Engineer

Aug 2025 - Dec 2025

Singapore Institute of Technology - Centre for Immersification, Singapore

Public Repository: [Link](#) | **Private Contribution Repository:** [Link](#)

- Developed R3CAP, an open-source platform that enables users to capture, reconstruct, edit, annotate, and collaboratively explore interactive 3D digital-twin environments across desktop and WebXR interfaces.
- Delivered full-stack features across the Python/FastAPI edge server and React/TypeScript/Babylon.js client, including scene authoring, spatial annotations, object and mesh workflows, shared-room synchronisation, MongoDB persistence, camera navigation, and browser-based 3D interaction.
- Extended the reconstruction pipeline through modular engine integration and asynchronous job orchestration across RTAB-Map, COLMAP, Blender, and experimental 3D Gaussian Splatting/Gaussian Grouping workflows, with status tracking, recovery handling, coordinate alignment, APIs, testing, and technical documentation supporting ACM CHI and ACM DIS submissions.

Section Leader - Police National Service Inspector

Nov 2024 - Aug 2025

Singapore Police Force - Operations Department, Contingency Planning Division, Singapore

Training: Officer Cadet Trainee, 80th Officer Cadet Course | Jun 2024 - Nov 2024

- Led personnel, task allocation, operational readiness, and cross-agency coordination for technology-enabled contingency-planning initiatives supporting major national operations, including GE2025 and NDP2025.
- Developed Python, UiPath RPA, Excel/VBA, ArcGIS, and dashboard workflows for data collection, validation, personnel accountability, logistics tracking, geospatial planning, and command reporting, saving an estimated 30+ analyst hours per operating cycle.

Cyber Security Engineer - Police National Service

Jan 2024 - Jun 2024

Home Team Science and Technology Agency - Home Team Security Operations Centre, Singapore

Training: Trainee Special Constable, Police Officers' Basic Course | Aug 2023 - Jan 2024

- Supported Security Operations Centre monitoring and incident response through SIEM alert triage, threat hunting, incident classification and escalation, digital-forensics investigations, evidence preservation, and repeatable response procedures.
- Automated recurring analyst reports and operational dashboards using Python and pandas; maintained security documentation and delivered cybersecurity demonstrations and briefings at the HTX TechX Summit and ministerial visits.

Workforce Data Insights Officer

Apr 2023 - Jul 2023

Ministry of Manpower, Singapore

- Conducted structured surveys, interviews, and stakeholder outreach for workforce-development, employment-policy, and labour-market research in accordance with government procedures, confidentiality requirements, and data-quality standards.
- Collected, validated, reconciled, and synthesised quantitative and qualitative workforce information, resolving inconsistent submissions and preparing accurate, analysis-ready records for internal reporting and policy research.

Artificial Intelligence and Data Analyst Intern

Jul 2022 - Feb 2023

Deloitte Consulting, Singapore

- Supported end-to-end analytics and consulting for a regional maritime client, spanning requirements analysis, client proposals, SQL querying, data validation, exploratory analysis, forecasting, model evaluation, and stakeholder deliverables.
- Built Python/pandas and SQL data workflows, engineered time-series features, and evaluated ARIMA, XGBoost, and LightGBM approaches; translated model findings into visualisations, recommendations, documentation, and client-facing presentations.

SELECTED PROJECTS

KithRelay - Tencent Cloud Hackathon 2026

2026

Stack: React, TypeScript, PDF.js, IndexedDB, Vitest | **Public Repository:** [Link](#)

- Co-developed a browser-local care-coordination workspace that parses PDFs and structured files into source-linked plans while keeping care documents on the user's device rather than uploading them to an application server.
- Engineered deterministic conflict reconciliation, mixed-patient and medication-status checks, evidence traceability, caregiver approval, Markdown exports, WorkBuddy folder integration, and 15 automated regression/UI tests.

Campus Mind - Alibaba Cloud x Qoder Hackathon 2026

2026

Stack: Python, FastAPI, Qwen, React/TypeScript, PyMuPDF, pytest, Docker | **Public Repository:** [Link](#)

- Co-developed a two-person active-recall platform that turns uploaded documents into Qwen questions, progressive hints, and source-grounded feedback through a React/TypeScript and FastAPI application.
- Implemented Pydantic validation, source checks, retry/fallback handling, EWMA mastery scoring, misconception tracking, Docker/Render delivery, and automated tests for ingestion, study flows, scoring, grounding, and AI failure modes.

BrainGit - Personal AI Assistant

2026

Stack: Python, FastAPI, Gemini, PostgreSQL/pgvector, OAuth, Docker | **Public Repository:** [Link](#)

- Architected a Telegram assistant with five specialised agents and 17 tools for email, calendars, tasks, and semantic notes, orchestrated through Gemini function calling and a bounded execution loop.
- Implemented pgvector semantic memory, asynchronous persistence, encrypted Google/Microsoft OAuth, access controls, scheduled briefings, Docker/Caddy deployment assets, and 27 mocked unit/integration tests.

IBM Telco Customer Churn - Machine Learning Classification

2026

Stack: Python, pandas, scikit-learn, imbalanced-learn, SMOTE | **Public Repository:** [Link](#)

- Co-developed a leakage-controlled study of 7,043 customers and 33 source features using imputation, scaling, encoding, SMOTE confined to training folds, stratified five-fold cross-validation, model comparison, and ablations.
- Selected L2 logistic regression and a 0.35 operating threshold, raising held-out recall from 55.3% to 88.5% with 46.4% precision, 0.608 F1, and 0.847 ROC-AUC while documenting the precision-recall trade-off.

SingHacks 2025 MSIG Travel Assistant

2025

Stack: Python, FastAPI, Streamlit, LangChain, Groq/Llama 3.3, PyMuPDF | **Public Repository:** [Link](#)

- Co-developed a conversational travel-insurance assistant using FastAPI, Streamlit, LangChain, and Groq-hosted Llama 3.3; authored chat endpoints, session memory, and intent/state routing for eligibility and policy-comparison workflows.
- Built deterministic comparison logic over three policy wordings and a PyMuPDF-to-LLM extraction pipeline for itinerary, passenger, and trip-cost context, constraining responses with a structured taxonomy and source-PDF links.

M5 Retail-Demand Forecasting Thesis and Major Project

2022 - 2023

Stack: Python, pandas, LightGBM, SARIMAX, XGBoost, Plotly | **Public Repository:** [Link](#)

- Completed an individual Temasek Polytechnic major project and forecasting thesis under Deloitte supervision using Walmart's hierarchical M5 dataset; engineered temporal features and reduced source-table memory by 67.8%-78.7%.
- Trained Tweedie LightGBM on 57.4 million rows with a 28-day temporal holdout, achieving local validation RMSE 3.497; authored a forecasting guide covering statistical and machine-learning models, multi-step strategies, feature selection, and evaluation.

EDUCATION

Bachelor of Science with Honours in Applied Artificial Intelligence | Singapore Institute of Technology | Expected 2028

Diploma in Financial Business Informatics | Temasek Polytechnic | 2023

CERTIFICATIONS & AWARDS

Certifications: RPA Developers Training Course - SimplifyNext (2025); CS50 - HarvardX/edX (2023); Python Data Science & Machine Learning Bootcamp - Udemy (2023); UiPath RPA Developer Foundation (2022)

Awards & Leadership: Best Operational Fitness / Vice-Chairperson - 80th OCC, SPF (2024); Best Trainee / Best Operational Fitness / Squad In-Charge - POBC, SPF (2023); Edusave Good Progress Award - MOE (2023); MUIS Madrasah Award, Secular Subjects (2017, 2020)